PREDICT & PROTECT

AI-Powered Accident Hotspot System

Moving from reactive reporting to predictive public safety.

THE PROBLEM & THE GOAL

THE CHALLENGE

Traffic accidents cause loss of life and significant economic damage. Traditional approaches mainly identify dangerous areas after accidents have occurred, making it difficult to proactively prevent collisions.

THE OBJECTIVE

Develop an AI-powered system that predicts where and when accident risks are highest, supporting preventative action and aiming to:

- Reduce collisions by **15%**
- Lower emergency response times
- Identify high-risk locations before accidents occur



HOW IT WORKS

1 DATA INGESTION

The system combines:

- Historical accident data
- Weather conditions
- Time of day
- Traffic density

2 AI ENGINE

The data is analysed using:

DBSCAN
Spatial Clustering
Identifies accident hotspots and geographic patterns.

LSTM
Time-Series Forecasting
Analyses historical patterns to forecast periods of increased accident risk.

3 OUTPUT

An interactive GIS heat map shows:

- Current high-risk zones
- Predicted future hotspots
- Geographic accident patterns

Used by city planners and emergency responders for **data-driven decisions.**

ACCIDENT RISK MAP



IMPACT & TECHNOLOGY

BENEFICIARIES

COMMUTERS
Safer roads and improved awareness of high-risk areas.

EMS
Better information for planning emergency resources and response.

CITY PLANNERS
Data-driven insights for road safety improvements and infrastructure planning.

TECH STACK

Python | TensorFlow | PostGIS | Mapbox

SUCCESS METRICS

85% Prediction accuracy

10% Faster emergency response

15% Reduction in collisions

100% Anonymised data (zero privacy breaches)